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Head-mounted display device

Abstract

A head-mounted display device includes a main body and a supporter. The main body has a connection portion. The supporter is pivotally connected to the connection portion by taking an axis as a rotation axis. There is a first gap between the supporter and the connection portion on one side away from a user's head. A width of the first gap remains constant during rotation of the supporter relative to the connection portion.

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Background/Summary

TECHNICAL FIELD

(1) The disclosure relates to a head-mounted display device, and more particularly, to a head-mounted display device with a supporter.

DESCRIPTION OF RELATED ART

(2) In recent years, due to the booming development of virtual reality (VR), augmented reality (AR) and mixed reality (MR) technologies, head-mounted display devices that utilize these

technologies are becoming increasingly popular in the market. Typical head-mounted display devices use a hinge to adjust the two sides of the device closer to or away from each other in order to adapt to the head size of different users. However, the exposed gap of the hinge of typical head-mounted display devices is too large and may change as the hinge rotates, which leads to unattractive appearance. Although there are also head-mounted display devices that cover the hinge with an appearance so that the hinge is then hidden, an additional appearance is required, which increases the manufacturing cost.

SUMMARY

(3) The disclosure provides a head-mounted display device that maintains an attractive appearance.

(4) The head-mounted display device of the disclosure includes a main body and a supporter. The main body has a connection portion. The supporter is pivotally connected to the connection portion by taking an axis as a rotation axis. There is a first gap between the supporter and the connection portion on one side away from a user's head. A width of the first gap remains constant during rotation of the supporter relative to the connection portion.

(5) Based on the above, in the head-mounted display device of the disclosure, the width of the first gap remains constant, thereby increasing the attractiveness of the appearance.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a schematic view of a head-mounted display device of an embodiment of the invention.

(2) FIG. 2 is a top sectional view of the head-mounted display device of FIG. 1 along a profile line A-A.

(3) FIG. 3A is a partial enlarged view of the vicinity of the connection portion of FIG. 2.

(4) FIG. 3B is a partial enlarged view of the supporter of FIG. 3A rotated to another position.

(5) FIG. 4A is a partial enlarged view of the vicinity of the connection portion of FIG. 2.

(6) FIG. 4B is a partial enlarged view of the supporter of FIG. 3A rotated to another position.

DESCRIPTION OF EMBODIMENTS

(7) FIG. 1 is a schematic view of a head-mounted display device of an embodiment of the invention. Referring to FIG. 1, the head-mounted display device **100** of the embodiment includes a main body **110**. When a user wears the head-mounted display device **100** on a user's head **50** with the main body **110** covering eyes of the user, the user may view the image displayed by the main body **110** conveniently. The image source of the head-mounted display device **100** may be a built-in display system or an external display device such as a smartphone, but the image source is not limited thereto. The image type displayed by the head-mounted display device **100** is, for example, virtual reality, augmented reality, or mixed reality, but the display type is not limited thereto.

(8) FIG. 2 is a top sectional view of the head-mounted display device of FIG. 1 along a profile line A-A. Referring to FIG. 1 and FIG. 2, in this embodiment, the head-mounted display device **100** further includes a supporter **160**. The main body **110** has a connection portion **130**. The supporter **160** is coupled to the connection portion **130** by taking an axis AX as a rotation axis. The number of the supporters **160** and the connection portions **130** of the embodiment is two, which are respectively located on the left and right sides of the head-mounted display device **100**, that is, on the left and right sides of the user's head **50**, but the disclosure is not limited thereto. By pivotally connecting the supporter **160** to the connection portion **130**, the two supporters **160** may be brought close to or away from each other, so as to adapt to the head size of different users. For example, if the user's head is relatively small, the two supporters **160** are brought closer to each other to better fit the user's head **50**, and vice versa.

(9) FIG. 3A is a partial enlarged view of FIG. 2. FIG. 3B is a partial enlarged view of FIG. 3A

rotated to another position. Referring to FIG. 3A, in this embodiment, there is a first gap G1 between the supporter **160** and the connection portion **130** on one side away from the user's head **50** (FIG. 2), and there is a second gap G2 between the supporter **160** and the connection portion **130** on one side close to the user's head **50**, while the first gap G1 and the second gap G2 are opposite to each other. In other words, the first gap G1 is located at the outer boundary of the supporter **160** and the connection portion **130**, while the second gap G2 is located at the inner boundary of the supporter **160** and the connection portion **130**.

(10) The axis AX is located on one side of the supporter **160** close to the connection portion **130** and is one side of the first gap G1. Therefore, the width of the first gap G1 remains constant during the rotation of the supporter **160** relative to the connection portion **130**. In other words, when the user adjusts the head-mounted display device **100**, the width of the first gap G1 remains constant.

(11) Since the head-mounted display device **100** does not use a typical hinge as a rotating mechanism, the supporter **160** may be extremely close to the connection portion **130**. That is, the width of the first gap G1 may be very small, i.e. the supporter **160** is closely adjacent to the connection portion **130**. As the width of the first gap G1 may be very small and remain constant when the supporter **160** is rotated, such a design allows the head-mounted display device **100** to maintain an attractive appearance and greatly reduce the possibility of pulling the user's hair.

(12) In this embodiment, the supporter **160** has a restricted segment **161**, a connection segment **162**, and a free segment **163**. The connection segment **162** is arc-shaped and connects the restricted segment **161** and the free segment **163**. For example, the supporter **160** is integrally formed, but the structure of the supporter **160** is not limited thereto.

(13) In this embodiment, the connection portion **130** has a chamber **131** and a chamber opening **132**, and the supporter **160** passes through the chamber **131** through the chamber opening **132**. When the supporter **160** passes through the chamber **131**, the restricted segment **161** is located in the chamber **131**. The connection segment **162** passes through the chamber opening **132** and is able to slide relative to the chamber opening **132**. The free segment **163** is exposed outside the chamber **131**.

(14) The size of the chamber **131** and the diameter of the chamber opening **132** may be adjusted according to design requirements, which are not limited in the disclosure.

(15) In this embodiment, when the connection segment **162** slides relative to the chamber opening **132**, the shapes of the connection segment **162** and the chamber opening **132** match with each other, so that the entire supporter **160** rotates around the axis AX.

(16) Specifically, referring to FIG. 3A and FIG. 3B, the chamber **131** has a first curved wall surface **140**, a second curved wall surface **142**, a first wall surface **144**, and a second wall surface **146**. The first curved wall surface **140** and the second curved wall surface **142** are opposite to each other and are located on two sides of the chamber opening **132** respectively. The first curved wall surface **140** is convex and is connected to the first wall surface **144** on one side away from the user's head **50** (FIG. 2). The second curved wall surface **142** is concave and is connected to the second wall surface **146** on one side close to the user's head **50** (FIG. 2). The shape of the first curved wall surface **140** and the shape of the second curved wall surface **142** correspond to each other.

(17) In this embodiment, the connection segment **162** is slidably disposed between the first curved wall surface **140** and the second curved wall surface **142**. The restricted segment **161** is limited between the first wall surface **144** and the second wall surface **146**. If the user's head is relatively big, the user may exert force on the supporter **160**. A part of the connection segment **162** then slides out of the chamber opening **132** along the first curved wall surface **140** and the second curved wall surface **142**, so that the free segment **163** of the supporter **160** moves outward. That is, by moving to the left side of the supporter **160** in FIG. 3A and FIG. 3B, the distance between the two supporters **160** is increased to better fit the large head size.

(18) When the restricted segment **161** contacts the second wall surface **146**, the second wall surface **146** stops the supporter **160**, and the supporter **160** stops rotating in time.

(19) Similarly, if the user's head is relatively small, the user may exert force on the supporter **160** in an opposite direction. The connection segment **162** then slides into the chamber opening **132** along the first curved wall surface **140** and the second curved wall surface **142**, so that the free segment **163** of the supporter **160** moves inward. That is, by moving to the right side of the supporter **160** in FIG. 3A and FIG. 3B, the distance between the two supporters **160** is decreased to better fit the small head size.

(20) When the restricted segment **161** contacts the first wall surface **144**, the first wall surface **144** stops the supporter **160**, and the supporter **160** stops rotating in time.

(21) FIG. 4A is a partial enlarged view of the vicinity of the connection portion of FIG. 2. FIG. 4B is a partial enlarged view of the supporter of FIG. 3A rotated to another position. In addition, referring to FIG. 4A and FIG. 4B, the radius of curvature of the first curved wall surface **140** is the radius of the bigger circle represented by the dotted line, while the radius of curvature of the second curved wall surface **142** is the radius of the smaller circle represented by the dotted line, and the two circles are centered on the axis AX. That is to say, the curvature centers of the first curved wall surface **140** and the second curved wall surface **142** are both located on the axis AX. On the other hand, the connection segment **162** matches the shapes of the first curved wall surface **140** and the second curved wall surface **142**, so the curvature center of the connection segment **162** is also located on the axis AX. In other words, the curvature centers of the connection segment **162**, the first curved wall surface **140**, and the second curved wall surface **142** of this embodiment are all located on the axis AX. In this way, the connection segment **162** can rotate smoothly between the first curved wall surface **140** and the second curved wall surface **142** without causing structural interference with the connection portion **130**.

(22) It should be noted that, conventional head-mounted display devices take a hinge as the rotation center, so as to make adjustment to fit the head shapes of different users. In contrast, the head-mounted display device **100** of this embodiment is slidably disposed between the two curved wall surfaces through the supporter **160**, taking the virtual axis as the rotation center without using any hinge, thereby achieving the effect of smooth rotation.

(23) Referring to FIG. 3A and FIG. 3B, on the other hand, the width of the second gap G2 gradually increases or decreases as the supporter **160** rotates relative to the connection portion **130**. The width of the second gap G2 has a minimum value when the restricted segment **161** is abutted against the first wall surface **144**. Conversely, the width of the second gap G2 has a maximum value when the restricted segment **161** is abutted against the second wall surface **146**.

(24) In this embodiment, the width of the second gap G2 is greater than zero, that is, the supporter **160** and the connection portion **130** do not contact each other inside the head-mounted display device **100**. Therefore, even if the user's hair falls in the second gap G2, the hair does not get pulled, which prevents the user from feeling pain or discomfort due to hair pulling. The user experience is thus enhanced.

(25) In addition, the minimum width of the second gap G2 of this embodiment is, for example, greater than or equal to the width of the first gap G1. Certainly, the size relationship between the width of the first gap G1 and the minimum width of the second gap G2 is not limited thereto, but depends on the design requirements.

(26) In addition, the second gap G2 is located inside the head-mounted display device **100**, that is, the side close to the user's head **50**. Therefore, when the user wears the head-mounted display device **100**, even if the width of the second gap G2 is greater than the width of the first gap G1, the second gap G2 is less likely to be seen and affect the appearance.

(27) Compared with conventional head-mounted display devices, the outer gap of the head-mounted display device **100** of the embodiment is extremely small and the width thereof remains constant during rotation, so that the overall attractive appearance is well maintained. Meanwhile, since the width of the outer gap and the minimum width of the inner gap of the head-mounted display device **100** are not zero, the user's hair is not pulled even if the user's hair falls in the outer

gap or the inner gap.

(28) In addition, in this embodiment, the material of the connection portion **130** is, for example, metal, but the material is not limited thereto as long as the connection portion **130** has sufficient structural strength. In this way, the chamber opening **132** is prevented from being deformed by the supporter **160**, causing the rotation axis of the supporter **160** to deviate from the preset axis AX. The width of the first gap G1 is also ensured to remain constant when the supporter **160** rotates, preventing the hair from being pulled.

(29) Based on the above, in the head-mounted display device of the disclosure, the width of the first gap remains constant during the rotation of the supporter relative to the connection portion. Therefore, the width of the first gap is extremely small and remains constant, thereby increasing the attractiveness of the appearance. In addition, when the width of the second gap is designed to be greater than zero, the user's hair is not pulled by the connection portion and the supporter, which prevents the user from feeling pain or discomfort due to hair pulling. The user experience is thus enhanced.

Claims

1. A head-mounted display device, comprising: a main body having a connection portion; and a supporter, pivotally connected to the connection portion by taking an axis as a rotation axis, wherein there is a first gap between the supporter and the connection portion on one side away from a user's head, and a width of the first gap remains constant during rotation of the supporter relative to the connection portion, and wherein an edge of the supporter is located on the axis, and the edge of the supporter remains positioned on the axis during rotation of the supporter relative to the connection portion.
2. The head-mounted display device according to claim 1, wherein the connection portion has a chamber and a chamber opening, and the supporter passes through the chamber through the chamber opening.
3. The head-mounted display device according to claim 2, wherein the chamber opening has a first curved wall surface and a second curved wall surface opposite to each other, the supporter has a connection segment, and the connection segment is slidably disposed between the first curved wall surface and the second curved wall surface.
4. The head-mounted display device according to claim 3, wherein the first curved wall surface is convex and is located on one side of the chamber away from the user's head.
5. The head-mounted display device according to claim 3, wherein a curvature center of the connection segment, the first curved wall surface, and the second curved wall surface are located on the axis.
6. The head-mounted display device according to claim 3, wherein the chamber has a first wall surface and a second wall surface, the first wall surface connects the first curved wall surface, the second wall surface connects the second curved wall surface, the supporter further comprises a restricted segment, and the restricted segment is limited between the first wall surface and the second wall surface.
7. The head-mounted display device according to claim 6, wherein the restricted segment connects the connection segment and is located in the chamber, there is a second gap between the supporter and the connection portion on one side close to the user's head, and a width of the second gap has a minimum value when the restricted segment is abutted against the first wall surface.
8. The head-mounted display device according to claim 6, wherein the restricted segment connects the connection segment and is located in the chamber, there is a second gap between the supporter and the connection portion on one side close to the user's head, and a width of the second gap has a maximum value when the restricted segment is abutted against the second wall surface.
9. The head-mounted display device according to claim 1, wherein the axis is located on one side

of the supporter close to the connection portion and is located on one side of the first gap.

10. The head-mounted display device according to claim 1, wherein there is a second gap between the supporter and the connection portion on one side close to the user's head, and a width of the second gap is greater than zero.
